Program: B Tech and MBA Tech Data Science, B Tech CSE (DS)	**Semester:** V/VI
Course: Virtualization and Cloud Computing	**Code:**

Teaching Scheme				**Evaluation Scheme**	
Lecture (Hours per week)	**Practical (Hours per week)**	**Tutorial (Hours per week)**	**Credit**	**Internal Continuous Assessment (ICA) (Marks - 50)**	**Term End Examinations (TEE) (Marks - 100)**
2	2	0	3	Marks Scaled to 50	Marks Scaled to 50

Pre-requisite: Database management system, Information security and privacy

Course Objective

To educate students on building cloud infrastructure based on a cloud computing reference model. It will focus on technologies, components, processes, and mechanisms of cloud infrastructure. It will educate students on various algorithms involved in better implementing the cloud-based systems. This course will enable students to take the EMC Cloud Infrastructure and Services Associate Certification (EMCCIS).

Course Outcomes

After completion of the course, students will be able to -

1. Describe the foundational concepts of computer networks and cloud computing, including network architectures, models, protocols, and cloud service/deployment models. (L1 – Remember)
2. Demonstrate the use of virtualization technologies and container tools such as hypervisors, Docker, and Kubernetes for cloud infrastructure. (L3 – Apply)
3. Implement and configure core AWS services including EC2, IAM, and S3 for secure and scalable cloud deployments. (L3 – Apply)
4. Analyze resource management techniques such as load balancing, auto scaling, and provisioning in distributed cloud systems. (L4 – Analyze)
5. Develop and deploy cloud-based applications using advanced AWS tools like Lambda, CloudFormation, SageMaker, and monitoring services. (L6 – Create)

Detailed Syllabus

Unit	Description	Duration
1.	Introduction to computer networks, Network Models, OSI Models and TCP/IP, Internet, Protocols, Network Applications	**03**
2.	**Introduction to Cloud** Distributed Computing and Enabling Technologies, Cloud Fundamentals: Cloud Definition, Evolution, Architecture, Applications, Essential characteristics of cloud computing, cloud deployment models, and cloud service models. Definition, Types of Virtualization, Hardware Virtualization, Full and Para Virtualization, Hypervisors, Hardware-assisted virtualization, operating system level virtualization, application virtualization, Containers, Docker and Kubernetes	**06**
3.	**Introduction to AWS** Various Platforms for Cloud Computing, Why AWS, Creating AWS Account, Basic Concept. **Identity and Access Management**	**03**

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	Overview, IAM Use cases and service details, IAM Users-Groups-Policies, MFA, AWS CLI setup, Roles, Security Tools	
4.	**Elastic Compute Cloud(EC2)** Basics Overview, Security Groups, SSH, EC2 Instance Connect. **Implementation** Study of Cloud computing Systems like Amazon EC2 and S3, Google App Engine, and Microsoft Azure, Build Private/Hybrid Cloud using open source tools.	**04**
5.	**Elastic Load Balancing (ELB) and Auto Scaling Groups(ASG)** High Availability, Scalability, Elasticity, Agility, Overview Applications for ELB and ALB, ASG along with hands-on **Resource Management and Load Balancing** Distributed Management of Virtual Infrastructures, Server consolidation, Dynamic provisioning and resource management, Resource Optimization, Scheduling, Load Balancing, various load balancing techniques.	**04**
6.	**S3, Databases and Analytics** Security Bucket Policy, Versioning, Website, Replication, Snowball, Snowball Edge and Snowmobile Overview **Databases** RDS and Aurora, Overview, ElastiCache, DynamoDb, RedShift Map Reduce and its extensions to Cloud Computing, HDFS, and GFS.	**04**
7.	**Other Compute Services** Lambda, ECS, Light sail, Cloud Formation, Route 53, Cloud Front, SNS and SQS, Cloud Watch, Cloud Trail, Amazon Elastic Beanstalk, Cognito, Sage Maker, Amazon SES, Amazon Polly, Transcribe, Customer carbon footprint tool, Amazon sustainability data initiative .	**06**
	Total	**30**

Text Books

1. Ian Foster and Dennis B. Gannon, *Cloud Computing for Science and Engineering*, 1st Edition, MIT Press, 2017.
2. Kevin L. Jackson, Scott Goessling, *Architecting Cloud Computing*, 1st Edition, Packt Publishing Limited, 2018.

Reference Books

1. Anthony T. Velte, *Cloud Computing: A Practical approach*, 1st Edition, Tata Mcgraw Hill Education Private Limited, 2017.
2. Marinescu D C, *Cloud Computing Theory and Practice*, 2nd Edition, Elsevier, 2017.
3. Andreas Wittig and Michael Wittig, *Amazon Web Services in Action*, manning publisher, 2018.
4. William Stallings, *Foundations of Modern Networking: SDN, NFV, QoE, IoT, and Cloud*, 1st Edition, Addison-Wesley Professional, Pearson, 2015.

Laboratory Work

8 to 10 programming exercises (and a practicum) based on the syllabus

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